Table of Contents

1. EXECUTIVE SUMMARY 4

1.1 Landslide risk assessment 4
1.2 Proposal of landslide mitigation measures 5

2. SECTION 1: LANDSLIDE RISK ASSESSMENT 6

2.1 Introduction 6
2.2 Available knowledge and required data 7
2.3 Step 1: Comprehensive multi-temporal landslide inventory 8
2.3.1 Objective 8
2.3.2 Method 8
2.3.3 Results 9
2.4 Step 2: Landslide susceptibility assessment 13
2.4.1 Objective 13
2.4.2 Method 13
2.4.3 Results 17
2.5 Step 3: Landslide hazard assessment 23
2.5.1 Objective 23
2.5.2 Method 23
2.5.3 Results 24
2.6 Step 4: Exposure and vulnerability 27
2.6.1 Objectives 27
2.6.2 Results 27
2.6.3 Comparative analysis of impacted land use categories 30
2.7 Step 5: Landslide risk assessment conclusion 31
2.7.1 Summary 31
2.7.2 Conclusion 31
2.8 Literature consulted for the landslide risk assessment 32
2.9 Appendices 37
2.9.1 Appendix 1 37

3. SECTION 2: PROPOSAL OF POTENTIAL LANDSLIDE MITIGATION MEASURES..... 39

3.1 Introduction 39
3.2 Types of mitigation measures 43
3.3 Criteria for selection of the mitigation measures 44
3.4 Measures to reduce the hazard 45
3.5 Measures to reduce the vulnerability 49
3.6 Measures to reduce the elements at risk 52
3.7 Highlight on CROM DSS 53
3.8 Gully erosion 55
3.9 Specific landslide mitigation measures for the study areas 57
3.9.1 Assumptions 57
3.9.2 Areas with the highest susceptibility to landslides 58
3.9.3 Proposed mitigation measures 59
3.9.4 Bill of quantities 60
3.10 Literature consulted for the landslide mitigation measures 60
3.11 Appendices 63
3.11.1 Appendix 1– landslide types and processes 63
3.11.2 Appendix 2 – mitigation approaches (extra information) 65
3.11.3 Appendix 3 – Examples of terracing types 67